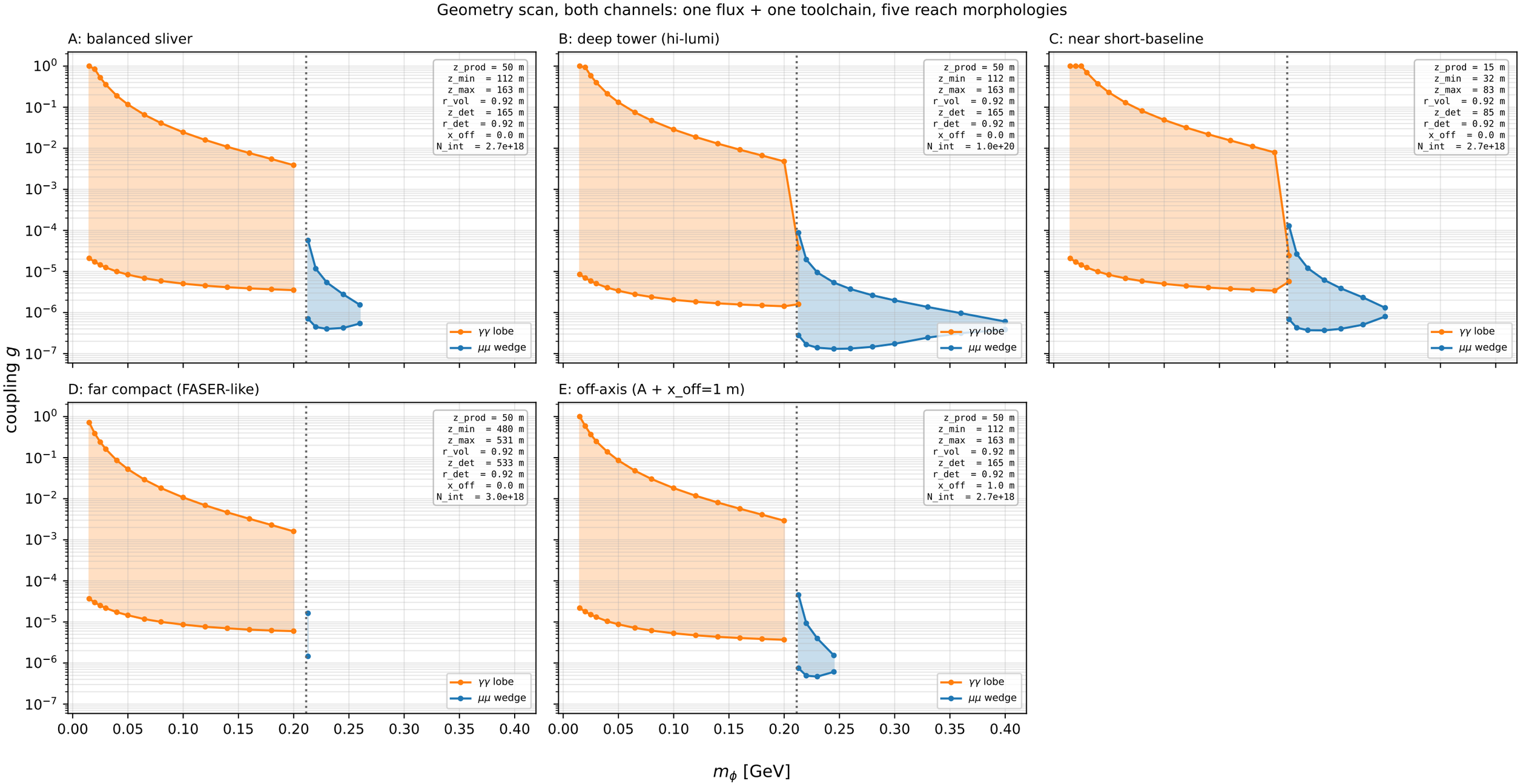
Geometry scan, both channels: one flux + one toolchain, five reach morphologies
A: balanced sliver
z_prod = 50 m
z_min = 112 m
z_max = 163 m
r_vol = 0.92 m
z_det = 165 m
r_det = 0.92 m
x_off = 0.0 m
N_int = 2.7e+18
B: deep tower (hi-lumi)
z_prod = 50 m
z_min = 112 m
z_max = 163 m
r_vol = 0.92 m
z_det = 165 m
r_det = 0.92 m
x_off = 0.0 m
N_int = 1.0e+20
C: near short-baseline
z_prod = 15 m
z_min = 32 m
z_max = 83 m
r_vol = 0.92 m
z_det = 85 m
r_det = 0.92 m
x_off = 0.0 m
N_int = 2.7e+18
D: far compact (FASER-like)
z_prod = 50 m
z_min = 480 m
z_max = 531 m
r_vol = 0.92 m
z_det = 533 m
r_det = 0.92 m
x_off = 0.0 m
N_int = 3.0e+18
E: off-axis (A + x_off=1 m)
z_prod = 50 m
z_min = 112 m
z_max = 163 m
r_vol = 0.92 m
z_det = 165 m
r_det = 0.92 m
x_off = 1.0 m
N_int = 2.7e+18
γγ lobe
μμ wedge
coupling g
m_ϕ [GeV]